

SmoothGrad: removing noise by adding noise

Authors: Daniel Smilkov, Nikhil Thorat, Been Kim,
Fernanda Viégas, Martin Wattenberg

Presented by: Vignav Ramesh, Zelin (James) Li, Paul Liu

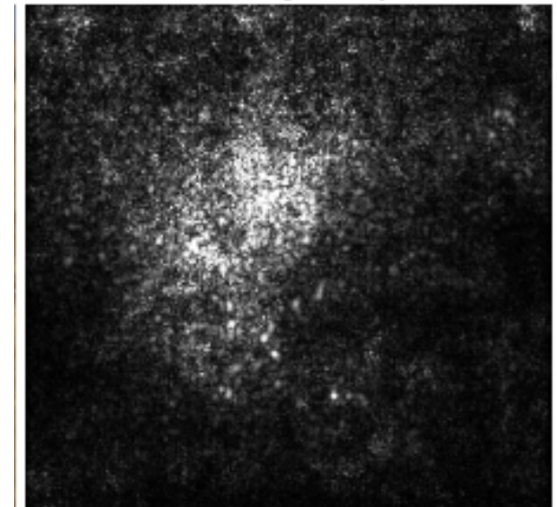
Motivation

- We want **post-hoc explanations of image classifiers**
- *Solution:* **Sensitivity maps** (a.k.a. saliency maps, pixel attribution maps)
 - Visual interpretation of gradient of class activation function w.r.t input image
 - Structured as grayscale image w/ dimension same as input image
 - Brightness of pixel $\propto$ importance to classification decision

Image



Sensitivity map M_c



Background: Gradients as Sensitivity Maps

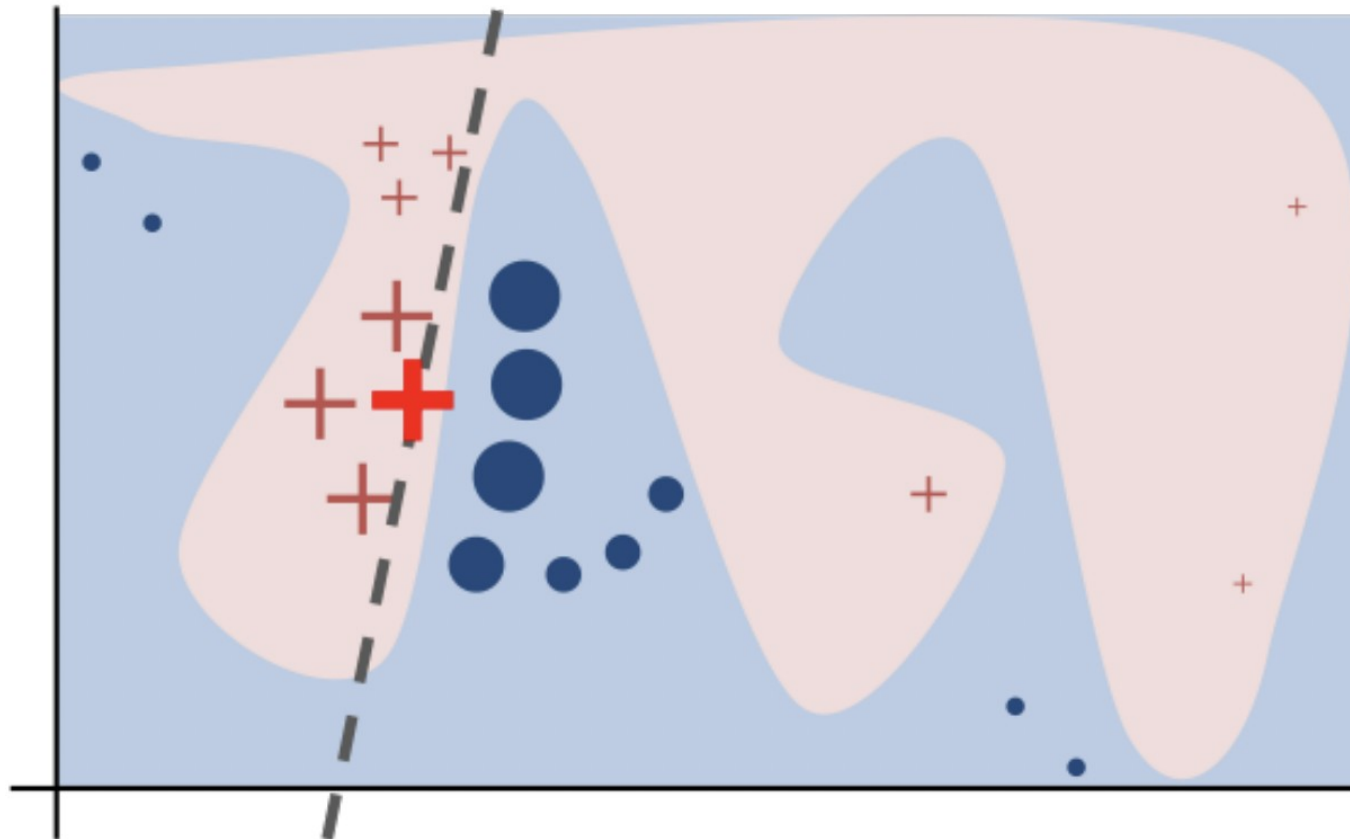
- N : network that classifies images into one class from set C
- Given input image x , N typically computes class activation function S_c for each $c \in C$
- Final classification $class(x) = \operatorname{argmax}_{c \in C} S_c(x)$
- **Sensitivity map given by**

$$M_c(x) = \partial S_c(x) / \partial x$$

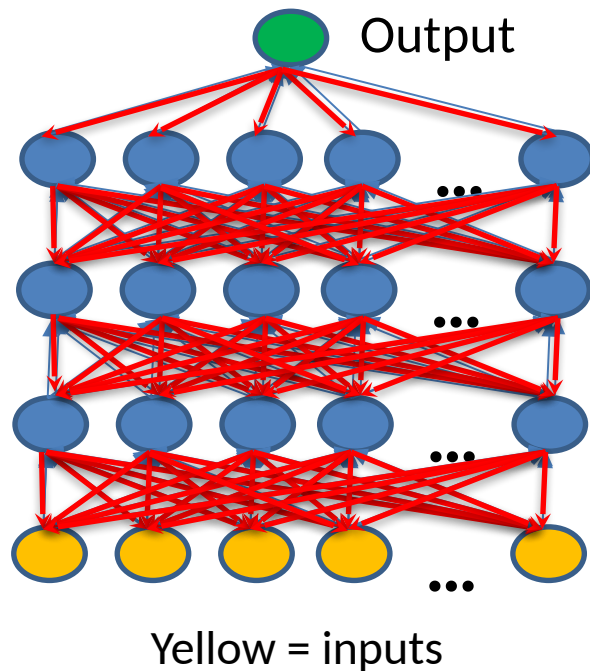
- **Intuition:** M_c represents how much difference a tiny change in each pixel of x would make to the classification score for class c

Related Work: Perturbation Methods

- **Key idea:** generate a perturbed dataset to fit an explainable model
 - LIME
 - KernelSHAP



Related Work: Backpropagation



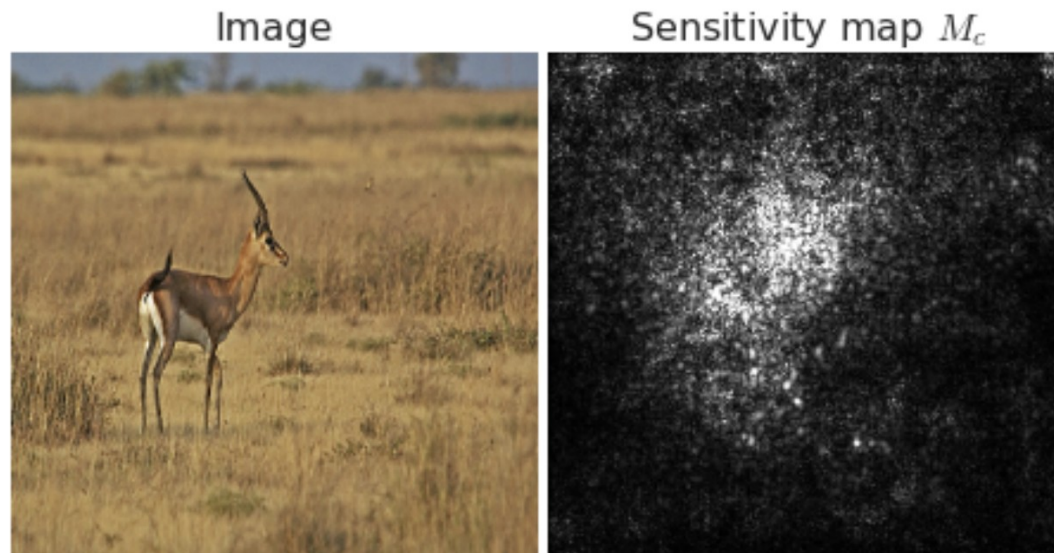
- **Key idea:** backpropagate importance through the network
 - Vanilla gradients
 - Layerwise relevance propagation (Bach et al.)
 - Integrated gradients (Sundararajan et al.)
 - DeepLIFT (Shrikumar et al.)
 - Deconvolution (Zeiler & Fergus, 2014)
 - Guided Backpropagation (Springenberg et al, 2014)

Limitations of Sensitivity Maps

■ Visually noisy

- Often highlight pixels that-to a human eye-seem randomly selected
- *a priori*, we cannot know if this noise reflects an underlying truth about how networks perform classification, or is due to more superficial factors

■ The Smoother
this soon!

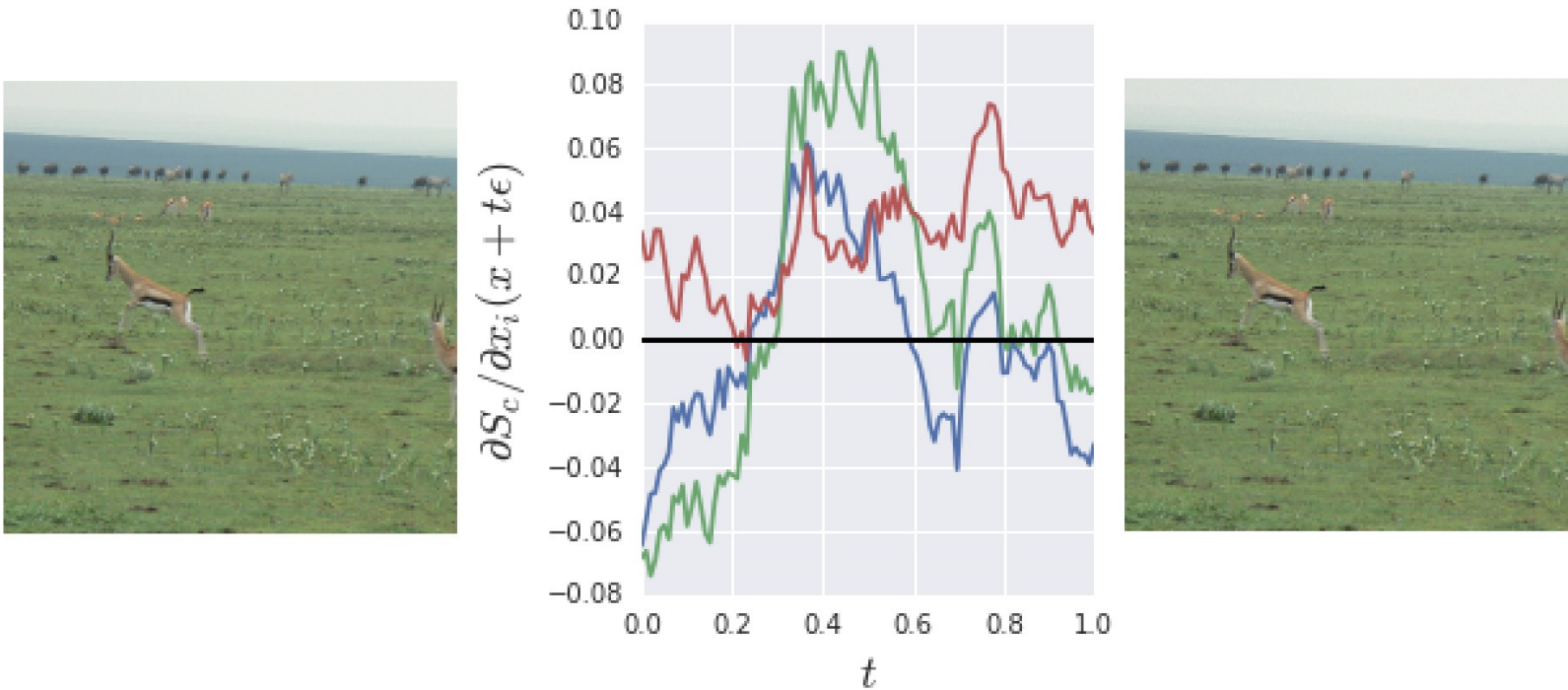


on - we'll get to

Theory Behind SmoothGrad: Noisy Gradients

- **Key idea behind SmoothGrad:** noisy maps are due to noisy gradients
- Derivative of S_c may fluctuate sharply at small scales
 - Apparent noise one sees in a sensitivity map may be due to essentially meaningless local variations in partial derivatives

Noisy Gradients (cont'd)



- Given these rapid fluctuations, gradient of S_c at any given point will be less meaningful than a local average of gradient values.

SmoothGrad: Intuition

- Recall that **noisy maps are due to noisy gradients**
- **Simple solution:**
 - take an image of interest
 - sample similar images by adding Gaussian noise to the image
 - take the average of the resulting sensitivity maps for each sampled image
 - This smoothes the gradient

SmoothGrad: Algorithm

1. Take random samples in a neighborhood of an input x with added noise
2. Average the result

$$\hat{M}_c(x) = \frac{1}{n} \sum_1^n M_c(x + \mathcal{N}(0, \sigma^2))$$

n is the number of samples, and $\mathcal{N}(0, \sigma^2)$ represents Gaussian noise with standard deviation σ .

Experimental Setup

- Performed SmoothGrad on visualizations of two neural networks:
 - Inception v3 model by Google that was trained on the ILSVRC-2013 dataset
 - Convolutional MNIST model based on the TensorFlow tutorial

Choosing Hyperparameters (σ : std. dev.)

$$\hat{M}_c(x) = \frac{1}{n} \sum_1^n M_c(x + \mathcal{N}(0, \sigma^2))$$

σ : the standard deviation
of the Gaussian noise

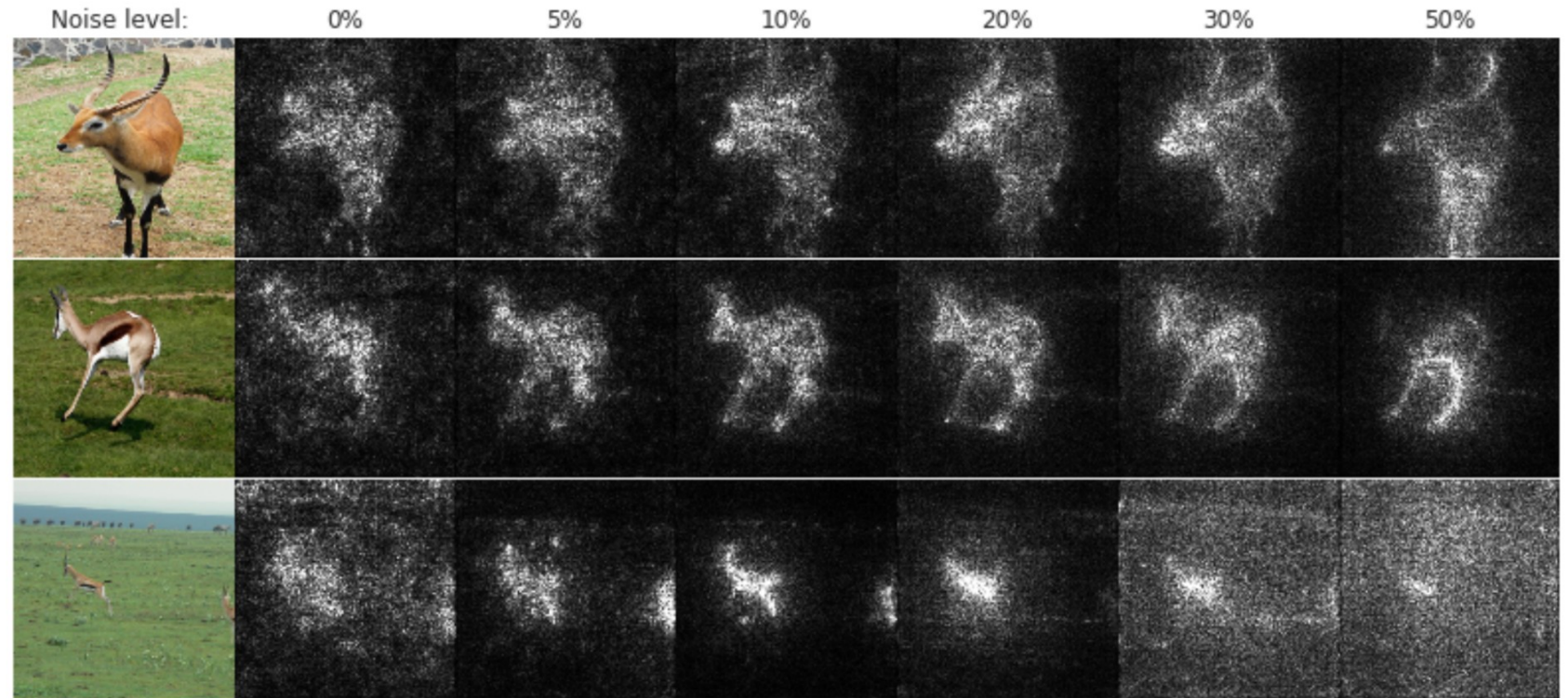


Figure 3. Effect of noise level (columns) on our method for 5 images of the gazelle class in ImageNet (rows). Each sensitivity map is obtained by applying Gaussian noise $\mathcal{N}(0, \sigma^2)$ to the input pixels for 50 samples, and averaging them. The noise level corresponds to $\sigma/(x_{max} - x_{min})$.

Choosing Hyperparameters (n : sample size)

$$\hat{M}_c(x) = \frac{1}{n} \sum_1^n M_c(x + \mathcal{N}(0, \sigma^2))$$

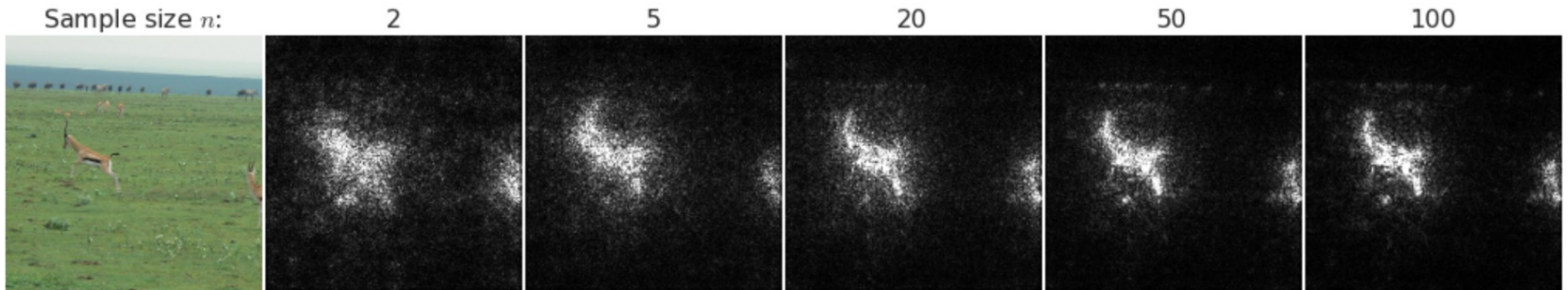


Figure 4. Effect of sample size on the estimated gradient for inception. 10% noise was applied to each image.

Qualitative Results: Visualization Techniques

- **Absolute Value of Gradients**

- depends on the characteristics of dataset



Qualitative Results: Visualization Techniques

- **Absolute Value of Gradients**

- depends on the characteristics of dataset

- **Capping outlying values**

- presence of few pixels that have much higher gradients than the average

- **Multiplying maps with images**

- capping to 99 percentile
- produce simpler & sharper images (Shrikumar et al., 2017; Sundararajan et al., 2017)
- Downside: Pixels with values of 0 will never show up on the sensitivity map.
- Upside: when viewing the importance of the feature as contribution to the image

Qualitative Results: Visual Coherence

Definition (Visual Coherence): Highlights are only on the object of interest, not the background

Comparison with three gradient-based methods

- Vanilla gradient
- Integrated Gradients
- Guided BackProp

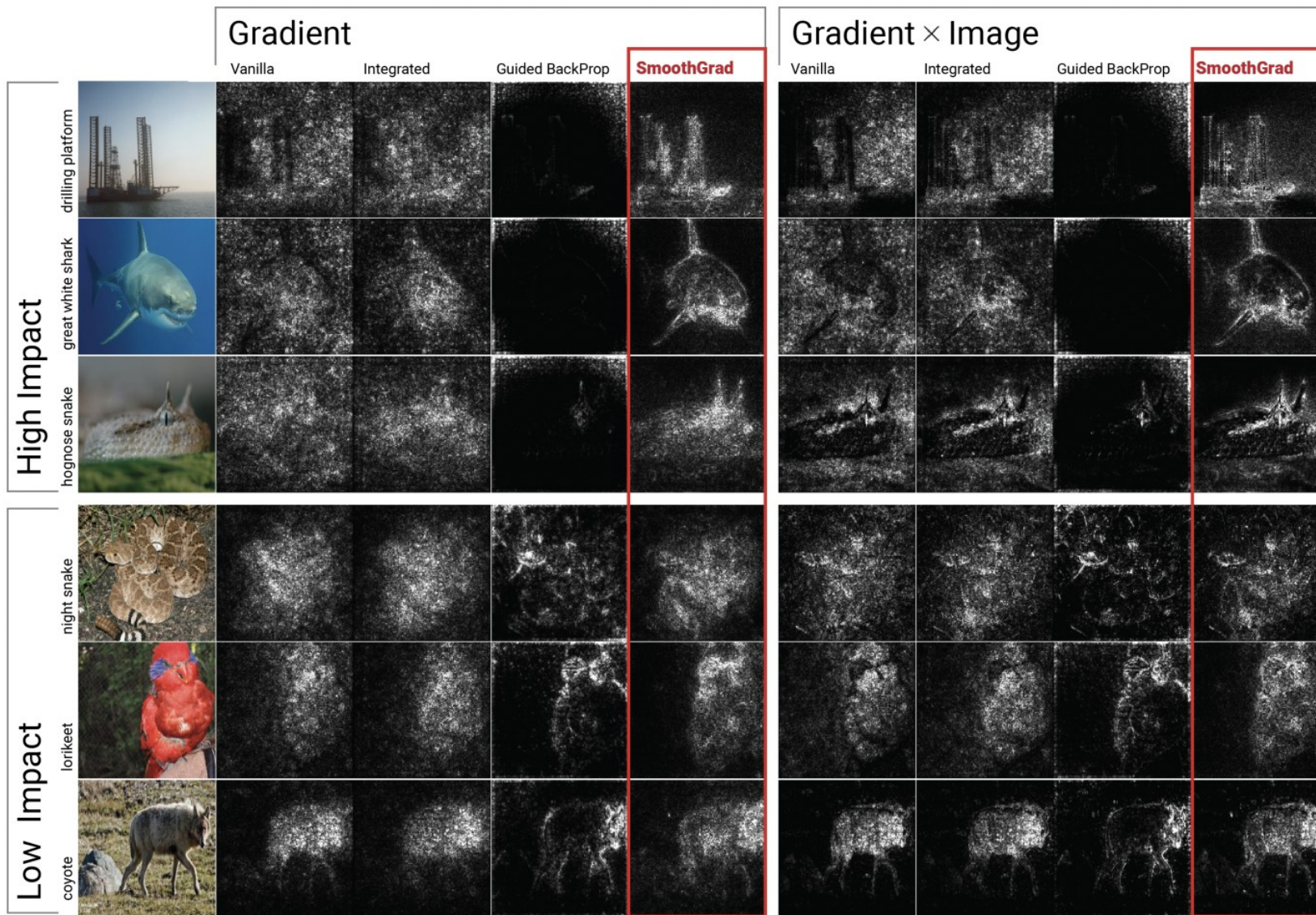


Figure 5. Qualitative evaluation of different methods. First three (last three) rows show examples where applying SMOOTHGRAD had high (low) impact on the quality of sensitivity map.

Qualitative Results: Discriminativity

Definition

(Discriminativity):
the ability to explain /
distinguish separate
objects without
confusion

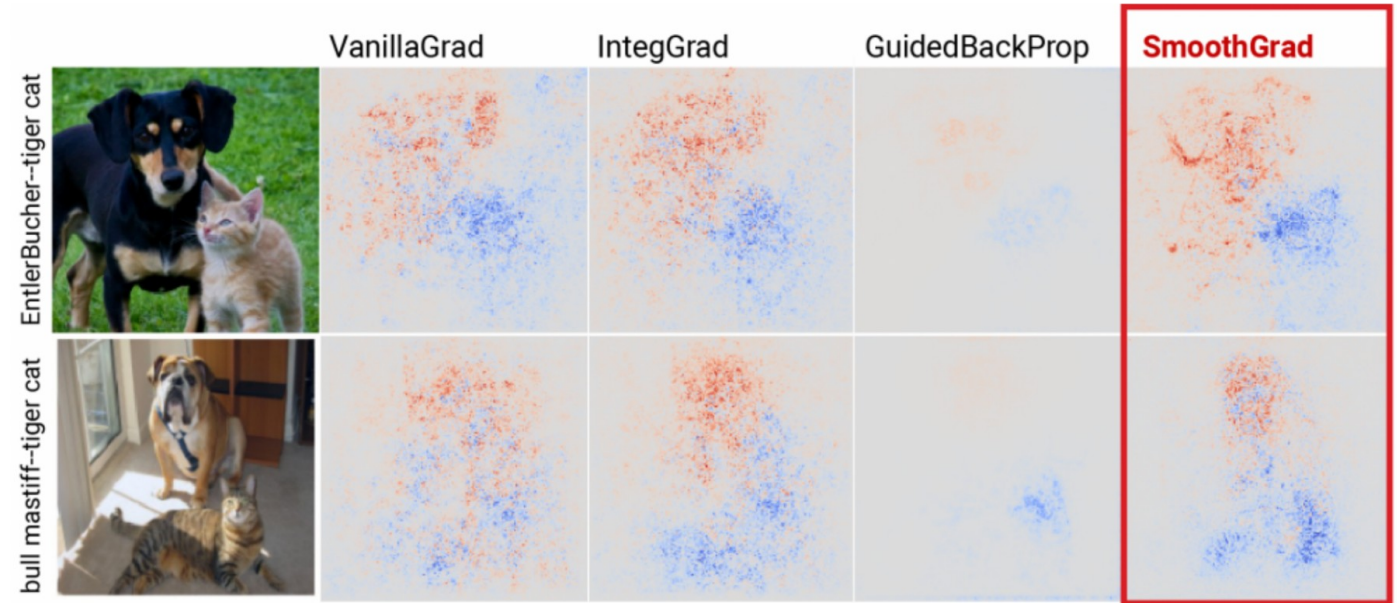


Figure 6. Discriminativity of different methods. For each image, we visualize the difference $\text{scale}(\partial y_1 / \partial x) - \text{scale}(\partial y_2 / \partial x)$ where y_1 and y_2 are the logits for the first and the second class (i.e., cat or dog) and $\text{scale}()$ normalizes the gradient values to be between $[0, 1]$. The values are plotted using a diverging color map $[-1, 0, 1] \mapsto [\text{blue}, \text{gray}, \text{red}]$. Each method is represented in columns.

Qualitative Results: Discriminativity

Open Problem

Which properties affect the discriminativity of a given methods?

- Why did GBP show the worst performance?

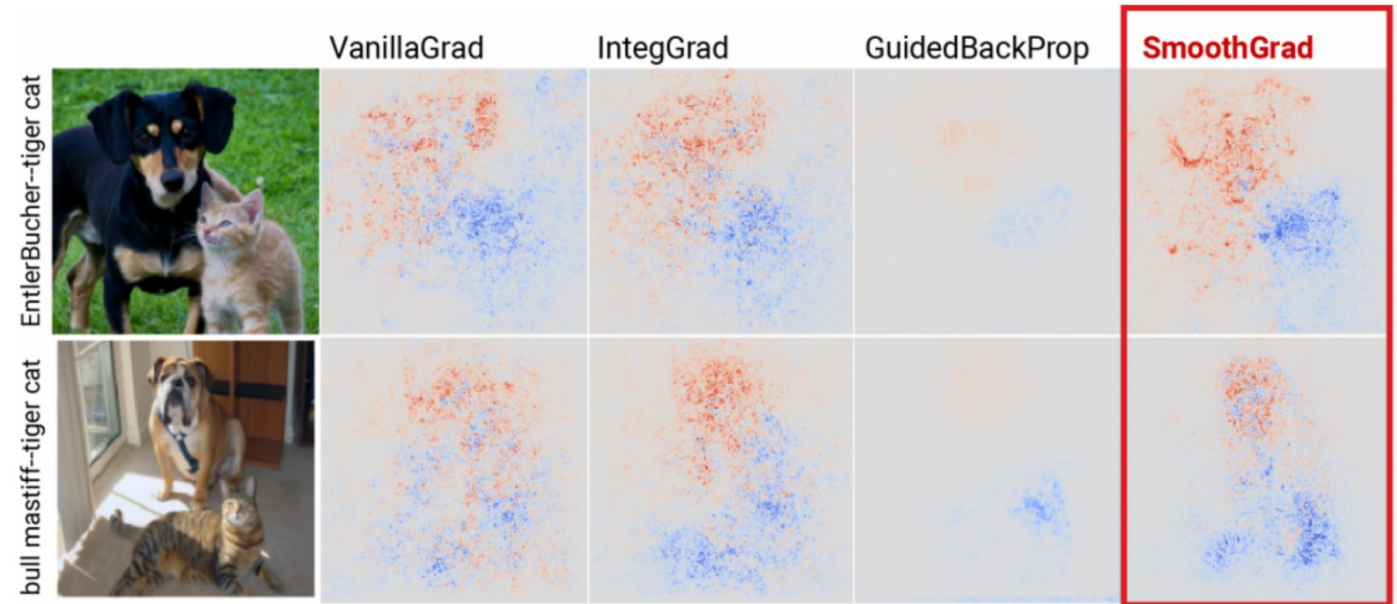
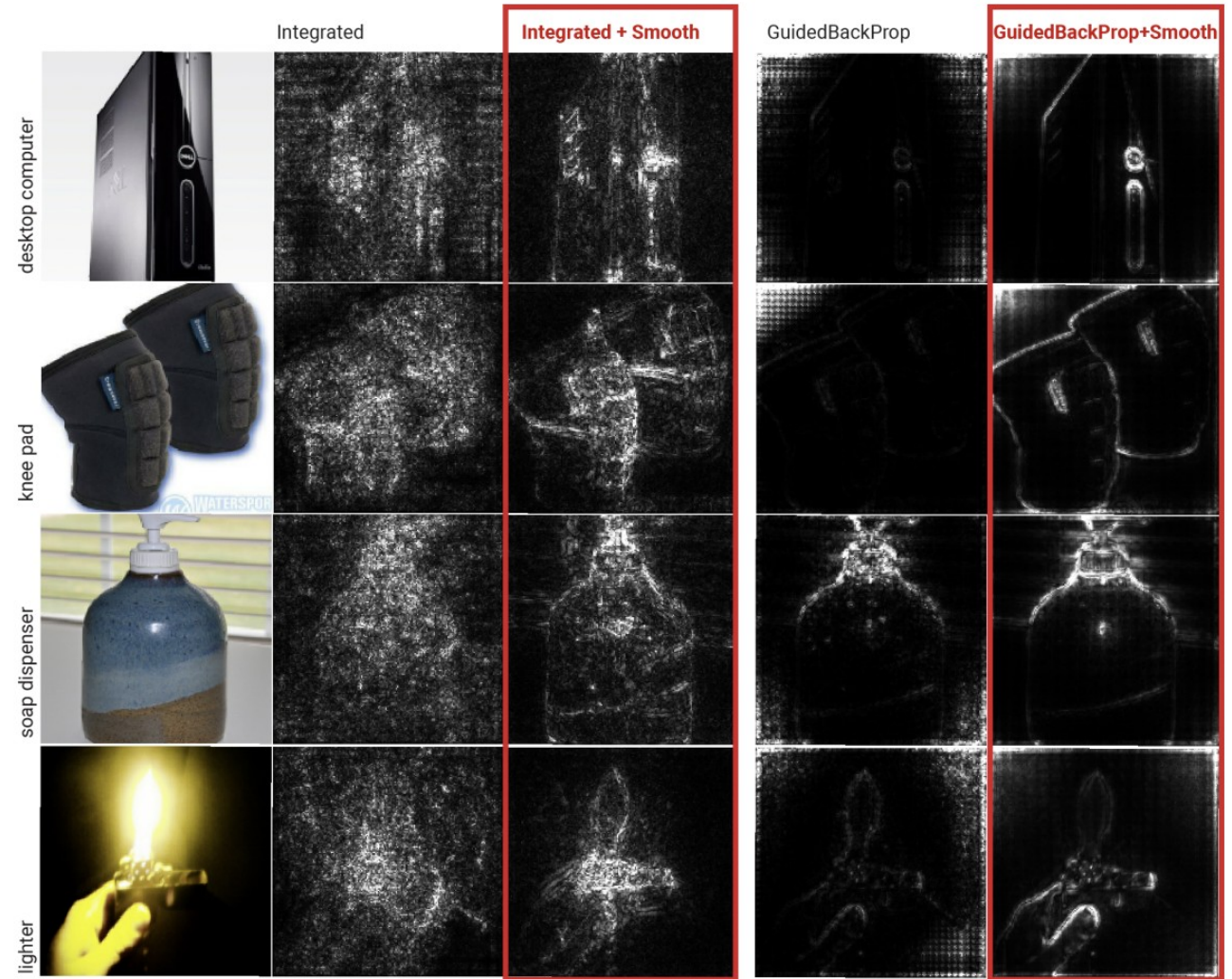


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Combining with Other Methods

The same smoothing procedure can be used to augment any gradient-based method.



Limitations / Discussion

- Completely qualitative results, can we get **quantitative metrics**?
- Noisy sensitivity maps are **due to noisy gradients**
 - Is this true?
 - Future work: look for further evidence and theoretical arguments
- Does SmoothGrad **generalize** to other networks & tasks?
- How do we tradeoff between making picture pretty and being faithful to the model? Do you think SmoothGrad handled this tradeoff well?



Thank you!

Questions?

Axiomatic Attribution for Deep Networks

Presented by Alex Lin, Steve Li, Kevin Huang

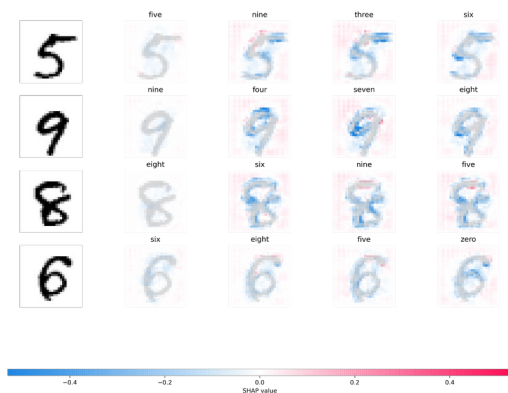
Mukund Sundararajan^{*1} Ankur Taly^{*1} Qiqi Yan^{*1}

Motivation and Problem Statement

- Feature Attribution:

Definition 1. Formally, suppose we have a function $F : \mathbb{R}^n \rightarrow [0, 1]$ that represents a deep network, and an input $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. An attribution of the prediction at input x relative to a baseline input x' is a vector $A_F(x, x') = (a_1, \dots, a_n) \in \mathbb{R}^n$ where a_i is the contribution of x_i to the prediction $F(x)$.

- Examples: in a CNN an attribution method could reveal which pixels were responsible for a certain label being picked (we saw this with LIME/SHAP)
- Problem: attribution technique are hard to evaluate empirically - hard to separate errors from model vs errors from attribution method
 - Ex. Gradients
 - Baseline: black image, empty text, etc.



Summary of Contributions

- Present two axioms: **Sensitivity** and **Implementation Invariance**
 - **Sensitivity:** For every input and baseline that differ in one feature but have different predictions then the differing feature should be given a non-zero attribution.
 - **Implementation Invariance:** The attributions are always identical for two functionally equivalent networks.
- 2 axioms → **integrated gradients**
 - Overview: path integral of the gradients along the straight line path from an input x to a baseline input x'

Two Axioms (Desiderata)

Sensitivity (a)

Definition: When 2 inputs that differ in only one feature result in different predictions, the **differing feature** should be given a **non-zero attribution**.

Invariance

Definition: The attributions are always identical for two functionally equivalent networks.

$$\frac{\partial f}{\partial g} = \frac{\partial f}{\partial h} \cdot \frac{\partial h}{\partial g}$$

Other Attribution Methods

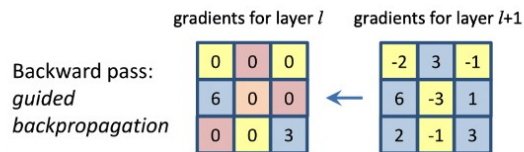
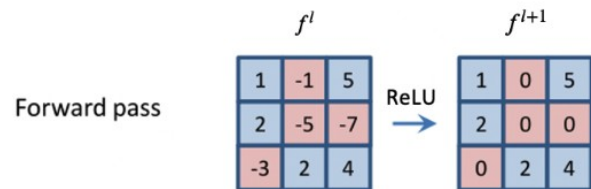
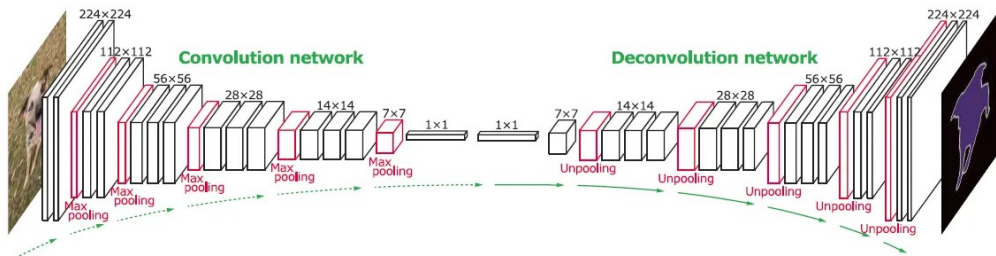
Gradients (of the output with respect to the input)

- Breaks sensitivity - prediction function can flatten at the input, giving 0 gradient despite function value at the input being different from the baseline
- Example:
 - Single ReLU network: $f(x) = 1 - \text{ReLU}(1 - x)$
 - Baseline: $x = 0$, input: $x = 2$
 - $f(0) = 0$, $f(2) = 1$
 - Since f is flat at $x = 1$, gradient gives attribution of 0 to x

Other Attribution Methods

Methods that Break Sensitivity

- DeConvNets, Guided back-propagation



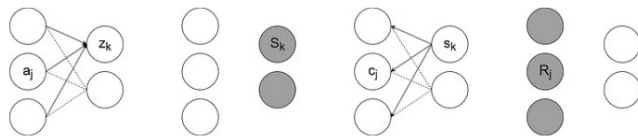
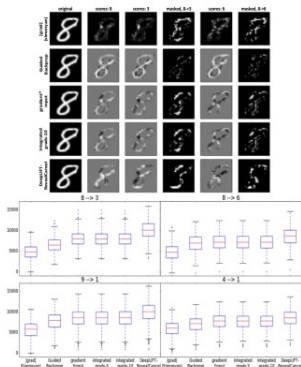
The gradients for both the red and yellow neurons are not backpropagated.

- Only back-prop through a ReLU if the ReLU is turned on at the input
 - Attribution is 0 for features with 0 gradients, despite non-zero gradient at the baseline

Other Attribution Methods

Methods that Break Implementation Invariance

- DeepLift and Layer-wise relevance propagation (LRP)



$$\forall_k : z_k = \epsilon + \sum_{0,j} a_j \cdot \rho(w_{jk}) \quad (\text{forward pass})$$

$$\forall_k : s_k = R_k / z_k \quad (\text{element-wise division})$$

$$\forall_j : c_j = \sum_k \rho(w_{jk}) \cdot s_k \quad (\text{backward pass})$$

$$\forall_j : R_j = a_j c_j \quad (\text{element-wise product})$$

- Replace gradients with discrete gradients, use a modified form of backpropagation
- Chain rule doesn't hold for discrete gradients (calculating gradients would be different) → breaks implementation invariance

The Method

Integrated Gradients

Definition

The **path integral** of the gradients along the **straight-line path** from the baseline x' to the input x .

$$\text{IntegratedGrads}_i(x) ::= (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{\partial F(x' + \alpha \times (x - x'))}{\partial x_i} d\alpha$$

New Axiom

Completeness: The sum of the attributions is equal to the difference of the outputs.

Proposition 1. *If $F : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable almost everywhere ¹ then*

$$\sum_{i=1}^n \text{IntegratedGrads}_i(x) = F(x) - F(x')$$

Uniqueness of Integrated Gradients

Path Methods

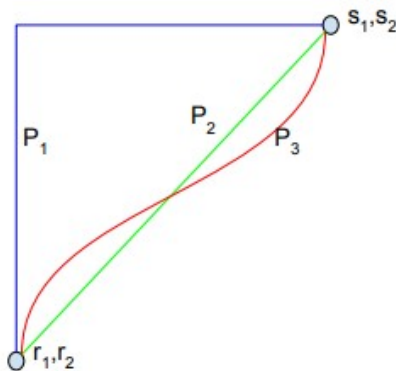


Figure 1. Three paths between an a baseline (r_1, r_2) and an input (s_1, s_2) . Each path corresponds to a different attribution method. The path P_2 corresponds to the path used by integrated gradients.

$$\text{PathIntegratedGrads}_i^\gamma(x) ::= \int_{\alpha=0}^1 \frac{\partial F(\gamma(\alpha))}{\partial \gamma_i(\alpha)} \frac{\partial \gamma_i(\alpha)}{\partial \alpha} d\alpha$$

Axioms

- **Sensitivity (b):** If the function does not depend (mathematically) on some input, then the attribution for that input is always zero.
- **Linearity:** Attributions preserve any linearity within the network.

$$a \times f_1 + b \times f_2$$

- **Symmetry-Preserving:** For symmetric variables, if they have identical values in the input and identical values in the baseline, they then receive identical attributions.

$$\text{Si} F(x, y) = F(y, x).$$

Using Integrated Gradients

Selecting a Baseline

Two Components:

- Zero-Score

$$F(x') \approx 0$$

- Conveys Absence of Signal

Examples:

- Object Recognition: All-black image
- Text: All-zero input embedding vector

Computing IGs

$$\text{IntegratedGrads}_i^{\text{approx}}(x) ::= (x_i - x'_i) \times \sum_{k=1}^m \frac{\partial F(x' + \frac{k}{m} \times (x - x'))}{\partial x_i} \times \frac{1}{m}$$

Experimental Results

Object Recognition CNN

Task: Given image, predict the category of the object

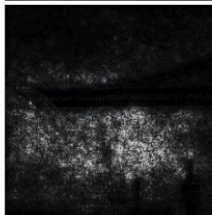
Image *Prediction* *Integrated Grad.* *Grad at Image*



Top label: reflex camera
Score: 0.993755



Top label: fireboat
Score: 0.999961



Top label: school bus
Score: 0.997033



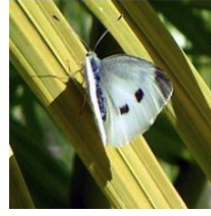
Image *Prediction* *Integrated Grad.* *Grad at Image*



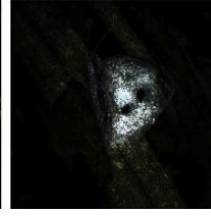
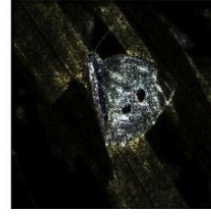
Top label: mosque
Score: 0.999127



Top label: viaduct
Score: 0.999994



Top label: cabbage butterfly
Score: 0.996838



Question Classification CNN

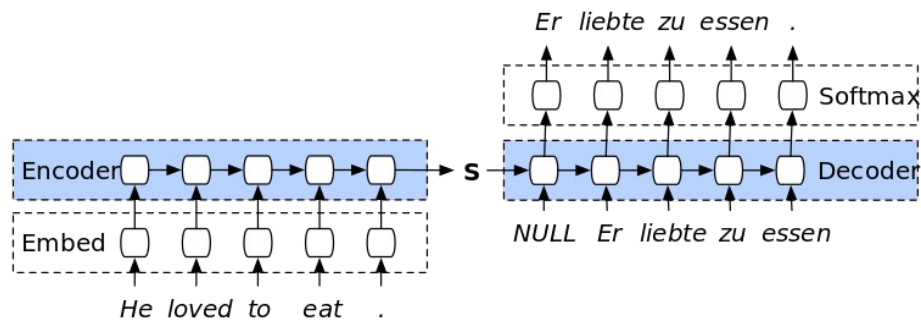
Task: Given question, predict what type of answer it is looking for.

how many townships have a population above 50 ? [prediction: NUMERIC]
what is the difference in population between fora and masilo [prediction: NUMERIC]
how many athletes are not ranked ? [prediction: NUMERIC]
what is the total number of points scored ? [prediction: NUMERIC]
which film was before the audacity of democracy ? [prediction: STRING]
which year did she work on the most films ? [prediction: DATETIME]
what year was the last school established ? [prediction: DATETIME]
when did ed sheeran get his first number one of the year ? [prediction: DATETIME]
did charles oakley play more minutes than robert parish ? [prediction: YESNO]

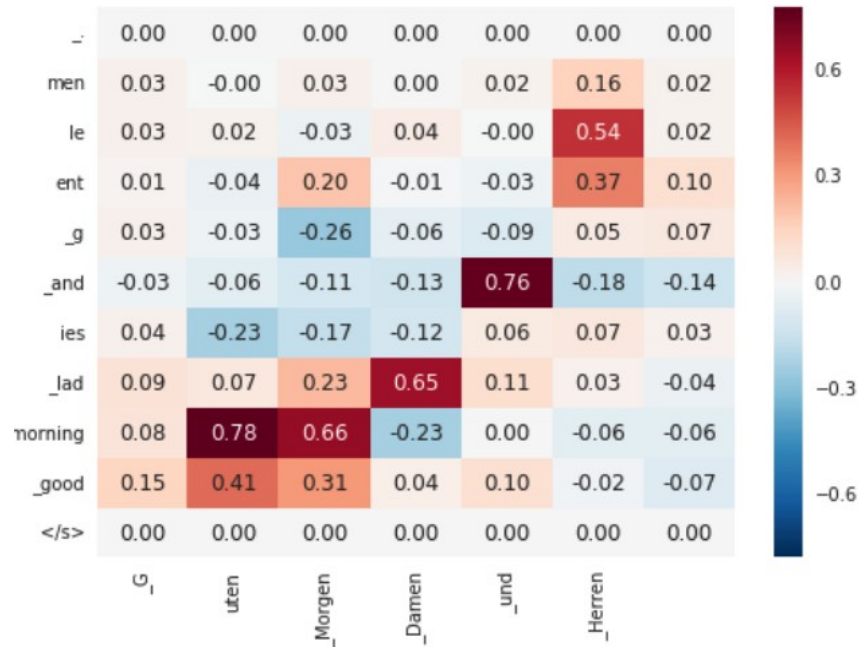
red = positive attribution, blue = negative attribution, gray = neutral attribution

Machine Translation RNN

Task: Given English sentence, predict German translation



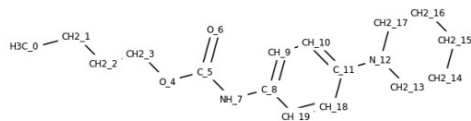
Example RNN Architecture



Ligand Screening Graph CNN

Task: Given molecular graph, predict whether it is active against an enzyme

Molecule: CID1562745



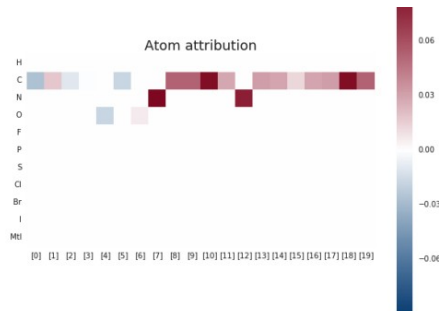
Attribution summary

Softmax score for task PCBA-588342: 0.98

Atom attribution: 0.62 (63%)

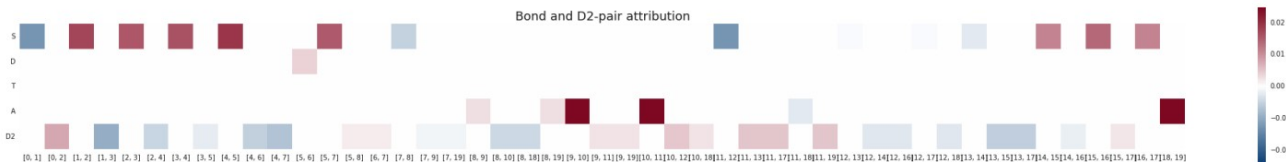
Bond attribution: 0.45 (46%)

D2-pair attribution: -0.03 (-3%)



Take-aways:

- More attribution to atom-pairs with bond (46%) compared to without bond (-3%)
- Attribution can help identify degenerate features (e.g. indicate that features are not fully convolved) (?)



Conclusion and Discussion

Summary

- Formalizes two axioms for attribution: **sensitivity, implementation invariance**
- Propose **integrated gradients** and argue that it is theoretically superior to other gradient-based methods (e.g. DeepLift, LRP, guided backprop, etc.)
- Perform experiments across several domains to showcase method

Discussion Questions

- Are you convinced that these axioms are desirable?
- Do you see any strengths or weaknesses in the idea of producing explanations through an integrated path?
- Have the experiments convinced you of the superiority of their method?